

Sampreet Patil

Belagavi, Karnataka | +91 7756062840 | samreetpatil270@gmail.com | sampreetpatil.com | [GitHub](#) | [LinkedIn](#)

SUMMARY

Recent Computer Science graduate specializing in AI-native applications, agentic systems, backend development, and scalable cloud architectures. Passionate about building reusable software IP using AWS cloud-native solutions.

EDUCATION

Visvesvaraya Technological University, Belagavi, Karnataka Graduated, June 2026
Bachelor of Technology in Computer Science and Business System CGPA: 8.56

EXPERIENCE

Angadi Technological Solutions Private Limited | *Backend Engineering Intern* November 2025 - Present

- Engineered high-throughput RESTful APIs and microservices for the "Enstine" online examination platform using Python (FastAPI) and Java (Spring Boot), optimizing end-to-end request latency.
- Architected containerized AWS cloud infrastructure, leveraging SQS for asynchronous message queueing, ECR and S3 for asset pipelines, and Lightsail for hosting—enforcing strict zero-trust access via IAM policies.
- Designed relational PostgreSQL database schemas and executed seamless data migrations via Alembic, maintaining database integrity across complex ERD models and high-concurrency transactions.

PROJECTS

Agent-X: An Agentic Personal Assistant | [GitHub](#) June 2025 - October 2025

Role: Founding Engineer & AI Architect

- Automated 10+ daily workflows with 85% accuracy by building a responsive, RAG-based agentic system.
- Engineered a scalable, AI-native microservices platform that delivers personalized news and maintains 99.5% uptime.
- Tech Stack:** Flutter, Python (FastAPI), Gemini LLM, RAG, ChromaDB, Firebase, Docker.

Pandas Assistant for Natural Data-Analytics | [GitHub](#) July 2025 – September 2025

Role: Full-Stack AI Engineer & Solution Architect

- Reduced data analysis time by 40% by translating natural language queries into real-time, interactive visualizations.
- Achieved 95% query accuracy by designing a modular AI architecture featuring robust, automated column mapping.
- Tech Stack:** Python (FastAPI, Pandas), React.js, OpenRouter API, WebSockets, Chart.js.

Crowd Detection with YOLOv5 and SORT | [GitHub](#) | [CV Expo Publication](#) Sep 2024 – Dec 2024

Role: Solo Developer & Computer Vision Engineer

- Achieved 95%+ real-time detection accuracy using intelligent frame sampling and a multi-threaded tracking system.
- Developed a production-ready GUI application featuring dynamic ROI density analytics and automated system alerts.
- Tech Stack:** Python, PyTorch, OpenCV, YOLOv5, SORT Algorithm, Tkinter.

SKILLS

Programming: Python, Java, Dart, C++, JavaScript, SQL, PostgreSQL

AI & Machine Learning: PyTorch, TensorFlow, LangChain, GenAI (RAG, Fine-Tuning), Pandas, NumPy, Scikit-Learn

Frameworks & Web: FastAPI, Flask, Flutter, React, Node.js

Tools & Platforms: Docker, Git, Postman, AWS, Prometheus, Grafana, Firebase, Clerk, Jupyter Notebooks

ACHIEVEMENTS AND ACTIVITIES

Codequest # 22 – National Coding Challenge (Naukri Campus) July 2025 – August 2025

6th Place | Solo Competition

- Ranked 6th nationally among 30,000 participants in a real-world algorithmic coding challenge.

CV Expo 2024 – Computer Vision Projects Showcase (Ready Tensor Inc.) Dec 2024 – Feb 2025

Featured Project | [Publication](#)

- Invited to present a solo-developed real-time crowd detection system using YOLOv5 and SORT at a global showcase of CV innovations.
- Invitation extended by Ready Tensor Inc. after review of GitHub project; recognized for practical application of computer vision and tracking systems.